UNIT 1

Activity: Product Purchase Data Analysis using Unix Command-Line Tools

CO Mapped: CO1 –Use Big Data and its Business Implications.

Type: Practical + Analytical Activity

Problem Statement

Organizations often store large volumes of transactional data in simple text files or log files. Use Unix command-line tools to compute summary statistics such as the total number of purchases, category-wise purchase counts, and most frequently purchased products.

Objective

This activity helps students understand how basic Unix command-line tools can be used for quick data exploration and analysis on structured text files.Students will develop skills in:

- Extracting specific columns from datasets
- Counting occurrences of data values
- Sorting and grouping textual data
- Identifying frequent patterns in datasets
- Performing quick data analysis using command pipelines

Product Purchase Data Analysis using Unix Command-Line Tools

1. Introduction

In modern organizations, large amounts of transactional data are generated and stored in simple text files such as logs or CSV datasets. Before performing advanced analytics using big data frameworks, it is often necessary to quickly explore and analyze the data to identify patterns and trends. Unix command-line tools provide powerful utilities for processing text-based datasets efficiently without requiring complex software or programming.

This activity demonstrates how basic Unix commands can be used to analyze a product purchase dataset stored in a text file. The goal of the analysis is to determine the total number of purchase records, examine sample data entries, analyze purchases by category, and identify the most frequently purchased products. These tasks help in understanding customer buying behavior and product popularity.

2. Dataset Description

The dataset used in this activity was stored in a file named **purchases.txt**. Each record in the dataset follows a simple structured format containing three fields separated by spaces:

(TransactionID ProductName Category)

Product Purchase Data Analysis using Unix Command-Line Tools

1. Dataset (purchases.txt)

T001 Laptop Electronics

T002 Mobile Electronics

T003 Rice Grocery

T004 Shirt Fashion

T005 Jeans Fashion

T006 Oil Grocery

T007 Laptop Electronics

T008 Mobile Electronics

T009 Rice Grocery

T010 Sugar Grocery

T011 Shirt Fashion

T012 Shoes Fashion

T013 Headphones Electronics

T014 Mobile Electronics

T015 Rice Grocery

T016 Laptop Electronics

T017 Oil Grocery

T018 Jeans Fashion

T019 Laptop Electronics

T020 Mobile Electronics

T021 Rice Grocery

T022 Sugar Grocery

T023 Shirt Fashion

T024 Shoes Fashion

T025 Laptop Electronics

T026 Mobile Electronics

T027 Rice Grocery

T028 Oil Grocery

T029 Jeans Fashion

T030 Laptop Electronics

T031 Mobile Electronics

T032 Rice Grocery

T033 Sugar Grocery

T034 Shirt Fashion

T035 Shoes Fashion

T036 Headphones Electronics

T037 Laptop Electronics

T038 Mobile Electronics

T039 Rice Grocery

T040 Oil Grocery

T041 Jeans Fashion

T042 Laptop Electronics

T043 Mobile Electronics

T044 Rice Grocery

T045 Sugar Grocery

T046 Shirt Fashion

T047 Shoes Fashion

T048 Laptop Electronics

T049 Mobile Electronics

T050 Rice Grocery

```
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ gedit purchases.txt
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ~/BDT_Activity3
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cat purchases.txt
T001 Laptop Electronics
T002 Mobile Electronics
T003 Rice Grocery
T004 Shirt Fashion
T005 Jeans Fashion
T006 Oil Grocery
T007 Laptop Electronics
T008 Mobile Electronics
T009 Rice Grocery
T010 Sugar Grocery
T011 Shirt Fashion
T012 Shoes Fashion
T013 Headphones Electronics
T014 Mobile Electronics
T015 Rice Grocery
T016 Laptop Electronics
T017 Oil Grocery
T018 Jeans Fashion
T019 Laptop Electronics
T020 Mobile Electronics
T021 Rice Grocery
T022 Sugar Grocery
T023 Shirt Fashion
T024 Shoes Fashion
T025 Laptop Electronics
T026 Mobile Electronics
T027 Rice Grocery
T028 Oil Grocery
T029 Jeans Fashion
T030 Laptop Electronics
T031 Mobile Electronics
T032 Rice Grocery
T033 Sugar Grocery
T034 Shirt Fashion
T035 Shoes Fashion
T036 Headphones Electronics
T037 Laptop Electronics
T038 Mobile Electronics
T039 Rice Grocery
T040 Oil Grocery
T041 Jeans Fashion
```

2. Unix Commands Used

Counting Total Purchase Records

The total number of purchase records was calculated using the wc -l command, which counts the number of lines in the dataset file.

Count total records: wc -l purchases.txt

Extracting and Analyzing Category Data

To analyze category-wise purchases, the cut command was used to extract the category column from the dataset.

Example command: cut -d' ' -f3 purchases.txt

The extracted category values were then sorted and counted using the sort and uniq -c commands.

Example command: cut -d' ' -f3 purchases.txt | sort | uniq -c

Identifying Most Frequently Purchased Products

To identify the most frequently purchased products, the product column was extracted using the cut command. The values were sorted and counted using uniq -c, and the results were sorted in descending order using sort -nr.

Example command: cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr | head –3

```
T038 Mobile Electronics
T039 Rice Grocery
T040 Oil Grocery
T041 Jeans Fashion
T042 Laptop Electronics
T043 Mobile Electronics
T044 Rice Grocery
T045 Sugar Grocery
T046 Shirt Fashion
T047 Shoes Fashion
T048 Laptop Electronics
T049 Mobile Electronics
T050 Rice Grocery
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ wc -l purchases.txt
50 purchases.txt
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d'' -f3 purchases.txt | sort | uniq -c
cut: the delimiter must be a single character
Try 'cut --help' for more information.
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f3 purchases.txt | sort | uniq -c

     20 Electronics
     13 Fashion
     17 Grocery
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr
      9 Rice
      9 Mobile
      9 Laptop
      5 Shirt
      4 Sugar
      4 Shoes
      4 Oil
      4 Jeans
      2 Headphones
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ head purchases.txt
T001 Laptop Electronics
T002 Mobile Electronics
T003 Rice Grocery
T004 Shirt Fashion
T005 Jeans Fashion
T006 Oil Grocery
T007 Laptop Electronics
T008 Mobile Electronics
T009 Rice Grocery
T010 Sugar Grocery
```

```
T007 Laptop Electronics
T008 Mobile Electronics
T009 Rice Grocery
T010 Sugar Grocery
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ head -5 purchases.txt
T001 Laptop Electronics
T002 Mobile Electronics
T003 Rice Grocery
T004 Shirt Fashion
T005 Jeans Fashion
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt
Laptop
Mobile
Rice
Shirt
Jeans
Oil
Laptop
Mobile
Rice
Sugar
Shirt
Shoes
Headphones
Mobile
Rice
Laptop
Oil
Jeans
Laptop
Mobile
Rice
Sugar
Shirt
Shoes
Laptop
Mobile
Rice
Oil
Jeans
Laptop
Mobile
Rice
Sugar
```

Display first few records: head purchases.txt

Display only Product column: cut -d' ' -f2 purchases.txt

Extract Category column: cut -d' ' -f3 purchases.txt

```
Rice
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f3 purchases.txt
Electronics
Electronics
Grocery
Fashion
Fashion
Grocery
Electronics
Electronics
Grocery
Grocery
Fashion
Fashion
Electronics
Electronics
Grocery
Electronics
Grocery
Fashion
Electronics
Electronics
Grocery
Grocery
Fashion
Fashion
Electronics
Electronics
Grocery
Grocery
Fashion
Electronics
Electronics
Grocery
Grocery
Fashion
Fashion
Electronics
Electronics
Electronics
Grocery
Grocery
Fashion
Electronics
```

```
Electronics
Grocery
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f3 purchases.txt | sort
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Electronics
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Fashion
Grocery
Grocery
Grocery
Grocery
Grocery
Grocery
Grocery
Grocery
```

Sort categories: cut -d' ' -f3 purchases.txt | sort

```
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' purchases.txt |sort | uniq -c
cut: you must specify a list of bytes, characters, or fields
Try 'cut --help' for more information.
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f3 purchases.txt | sort | uniq -c
     20 Electronics
     13 Fashion
     17 Grocery
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt
Laptop
Mobile
Rice
Shirt
Jeans
Oil
Laptop
Mobile
Rice
Sugar
Shirt
Shoes
Headphones
Mobile
Rice
Laptop
Oil
Jeans
Laptop
Mobile
Rice
Sugar
Shirt
Shoes
Laptop
Mobile
Rice
Oil
Jeans
Laptop
Mobile
Rice
Sugar
Shirt
Shoes
```

Count purchases per category: cut -d' ' -f3 purchases.txt | sort | uniq -c

Most frequent products: cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr

Top 3 products: cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr | head –3

```
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt | sort | uniq -c
      2 Headphones
      4 Jeans
      9 Laptop
      9 Mobile
      4 Oil
      9 Rice
      5 Shirt
      4 Shoes
      4 Sugar
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr
      9 Rice
      9 Mobile
      9 Laptop
      5 Shirt
      4 Sugar
      4 Shoes
      4 Oil
      4 Jeans
      2 Headphones
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$ cut -d' ' -f2 purchases.txt | sort | uniq -c | sort -nr | head -3
      9 Rice
      9 Mobile
      9 Laptop
hduser@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~/BDT_Activity3$
```

5. Challenges Faced

Several challenges were encountered during the implementation of this activity. One challenge was ensuring that the dataset followed a consistent format so that column extraction commands could work correctly. Another issue involved understanding how to combine multiple Unix commands using pipelines to perform complex operations such as sorting and counting data values.

Additionally, careful attention was required to correctly specify column delimiters and field numbers when using the cut command. These challenges were resolved by testing commands step by step and verifying intermediate outputs.

6. Conclusion

This activity demonstrates how Unix command-line tools can be used for quick and efficient analysis of structured text data. Commands such as cut, sort, uniq, wc, and head allow users to extract columns, count occurrences, and identify patterns in datasets with minimal effort. These tools are widely used in data engineering and big data preprocessing because they enable fast exploration of large datasets before applying advanced analytics techniques.